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Technology - European Semiconductors | Europe

AI GPU/ASIC Tracker, China LLMs, Advanced Packaging and MLCC - The Next AI Bottleneck

Join technology analysts Shawn Kim, Charlie Chan, Nigel Van Putten, Daisy Dai, Lydia Lin today to discuss recent developments and investment implications in the AI semiconductor and hardware supply chains, advanced packaging, power semis and MLCC. Register [here](#).

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TECHNOLOGY - EUROPEAN SEMICONDUCTORS	
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Industry View	In-Line

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OVERVIEW

Global Technology Webcast
AI GPU/ASIC Tracker, China LLMs, Advanced Packaging and MLCC – The Next AI Bottleneck



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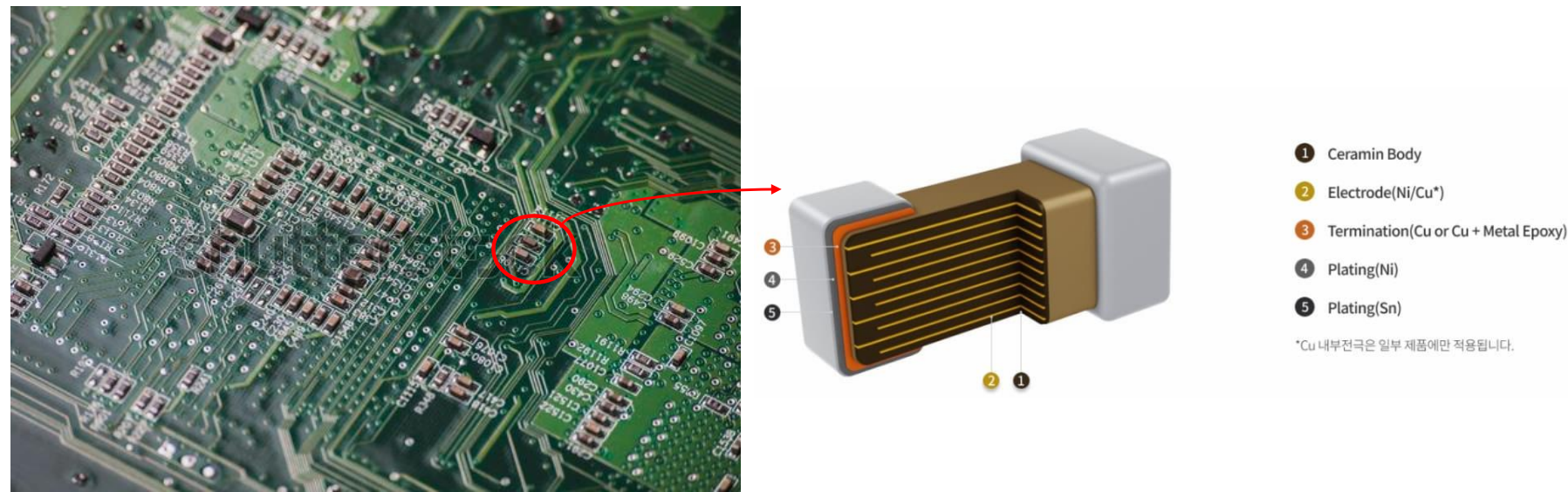
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Head of Asia and Europe Technology Research

MLCCs: Small Components That Enable Modern Electronics

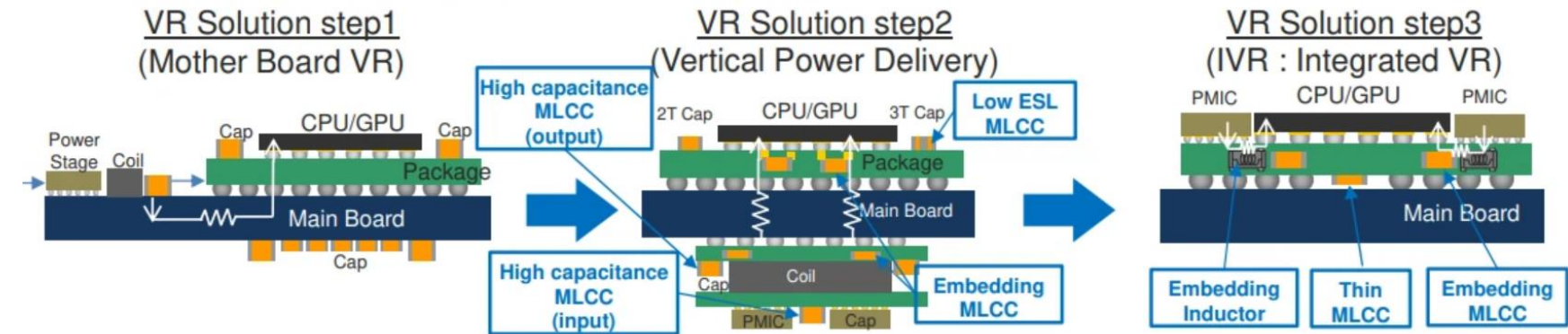
MLCCs (Multi-Layer Ceramic Capacitors) are essential passive components used for power stabilization, noise filtering, and signal integrity.



Source: Shutterstock, Samsung Electro-Mechanics

AI Servers Require More MLCCs as Power Delivery Architectures Evolve

New server designs require higher-capacitance, lower-ESL, and embedded MLCC solutions positioned closer to CPUs and GPUs.



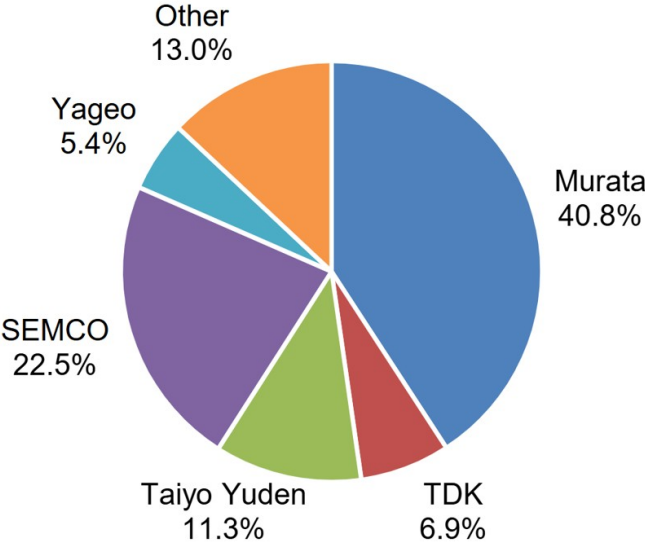
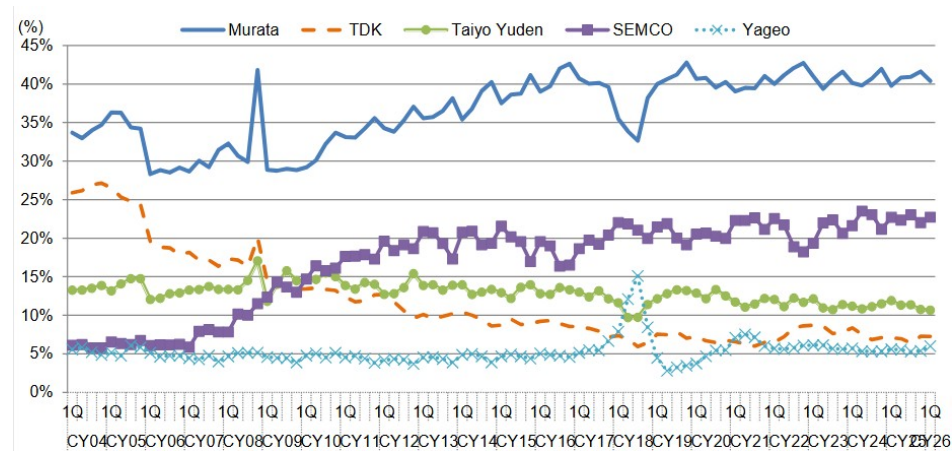
Source: Taiyo Yuden.

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The MLCC Industry Is Highly Consolidated

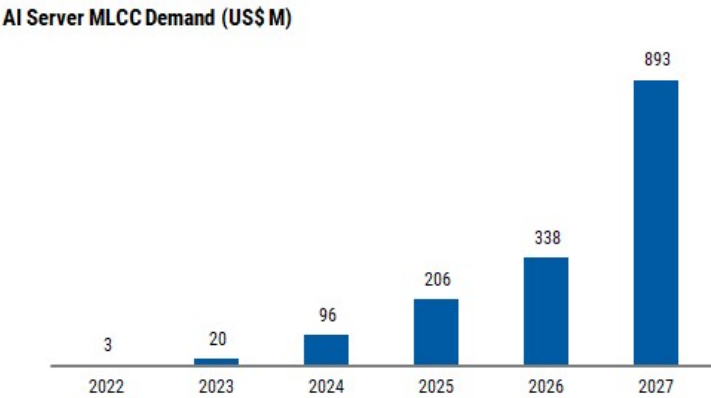
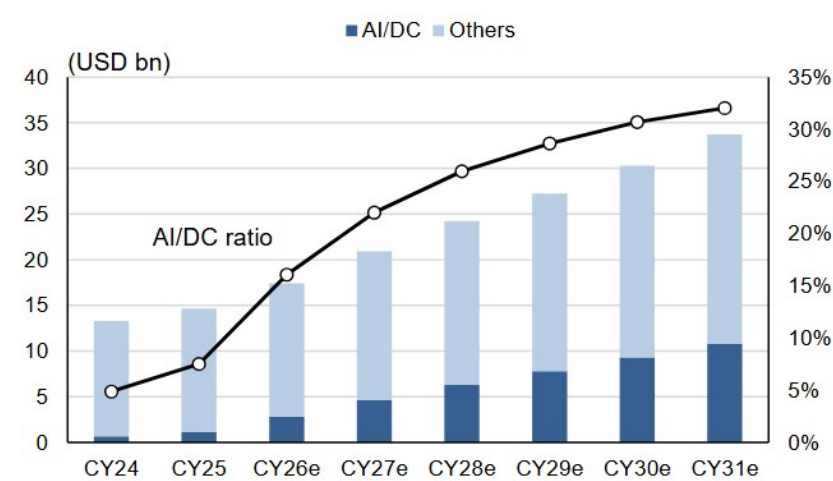
The top five suppliers account for ~87% of the global MLCC market (CY25), led by Murata (OW) and SEMCO (OW).



Source: Company data, Morgan Stanley Research.

AI Is Emerging as a Key Growth Driver for the MLCC Industry

We estimate AI server MLCC demand to approach US\$1bn by 2027 driven by content growth and spec upgrade.



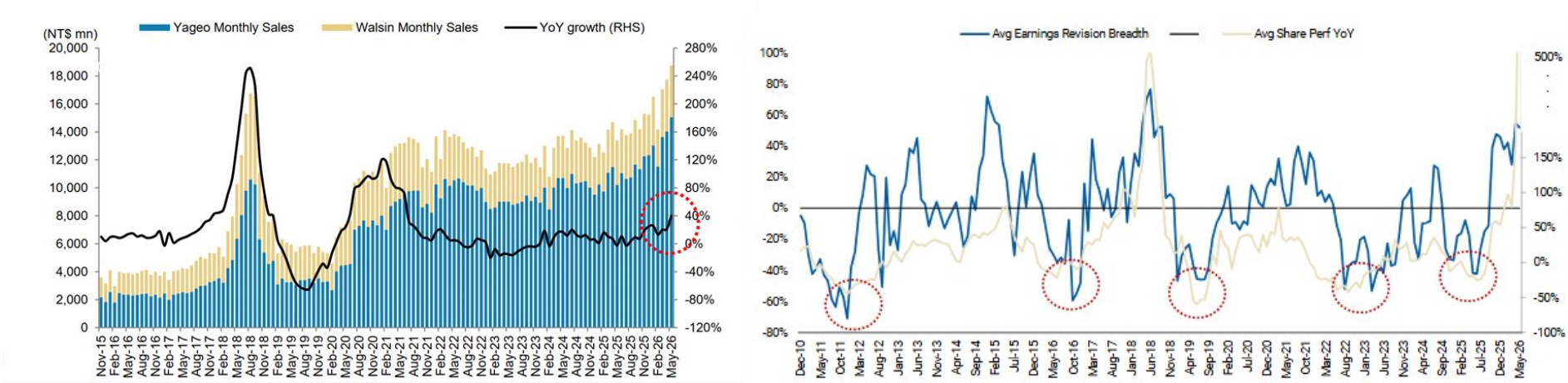
Source: Company data, Morgan Stanley Research.

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MLCC Fundamentals Are Inflecting

Monthly sales growth among key Taiwan MLCC suppliers has recently turned positive. Earnings revision breadth has improved meaningfully since late.



Source: Factset, Morgan Stanley Research estimates.

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Charlie Chan

Head of Greater China Semiconductor Research

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AI Semis: Now and the Future – “Prompts”

- Generative AI Demand Driver:
 - Killer apps
 - Competition
 - Sovereign
- Growth limitations:
 1. Budget
 2. Energy -> in the US
 3. Chip Capacity -> in China
 4. Regulation
- Semiconductor solutions:
 - Moore's Law
 - CoWoS/ SoIC
 - HBM
 - CPO
 - Custom chip
 - GaN HVDC 800V
- Growth perspective:
 1. Inference vs. Training AI semis
 2. Edge vs. Cloud AI semis
 3. Custom ASIC vs. AI GPU

CSPs Still Need Custom Chips Even with NVIDIA (OW) Providing Powerful AI GPUs

Global CSP cloud AI custom chips

Google TPU entering 6th generation



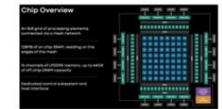
AWS AI training solution Trainium



AWS AI training solution Inferentia



Meta MTIA v1 with RISC-V cores



Habana developed the Gaudi chip



Tesla is introducing Dojos



Latest evidence of AI ASIC: AWS Trainium3



Trainium Forecasts from MS Asia Research team

k Units	2023	2024	2025e	2026e	2027e	2028e
Trainium1/Inferentia2	300	600	30			
Trainium2		300	1,520			
Trainium3			20	1,600	1,200	-
Trainium4					600	2,000
Total	300	900	1,570	1,600	1,800	2,000

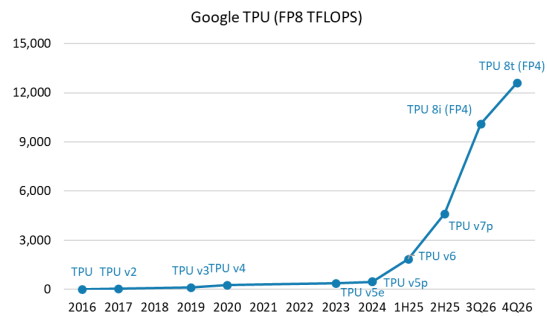
Source: AWS, Google, Meta, Intel, Tesla, Morgan Stanley Research (e) estimates.

More ASIC Projects Are Coming, According to Each CSP Plan

TPU Forecasts from MS Asia Research team

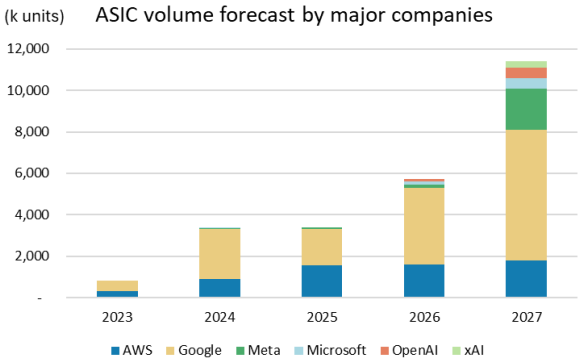
k Units	2023	2024	2025e	2026e	2027e	2028e
v5	500	2,400	250			
v6 (Trillium)			1,000			
v7 (Ironwood, by Broadcom)			500	2,300	500	
v8i (Sunfish; 3nm, by Broadcom)				900	3,000	2,500
v8t (Zebrafish; 3nm, by MediaTek)				500	2,500	1,000
v9 (Humufish; 2nm, by MediaTek)					150	2,500
v9a (Merope; 2nm, by US design service)						unknown
v10 (Icefish; 1.4nm, by MediaTek)						unknown
Total	500	2,400	1,750	3,700	6,150	>6000

Google TPU performance is growing rapidly



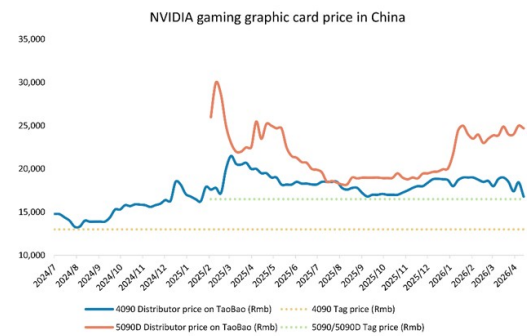
Source: Company data, Morgan Stanley Research (e) estimates

ASIC volume forecast by major companies

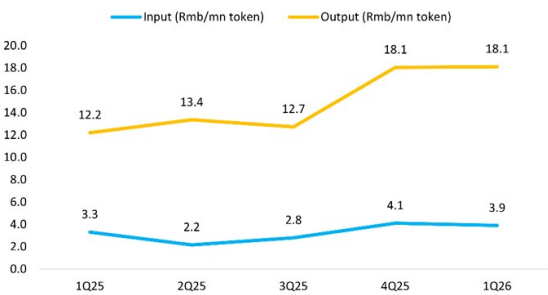


Near-term market tracker for China’s AI GPU demand

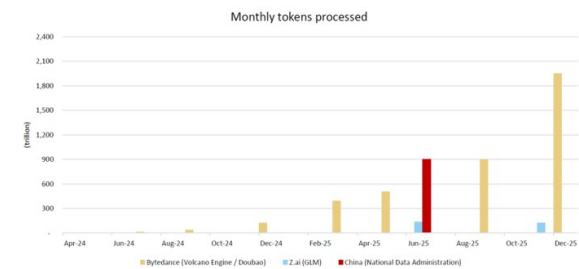
Nvidia's 5090 price keeps rising in China



Average token price for China's mainstream AI LLMs

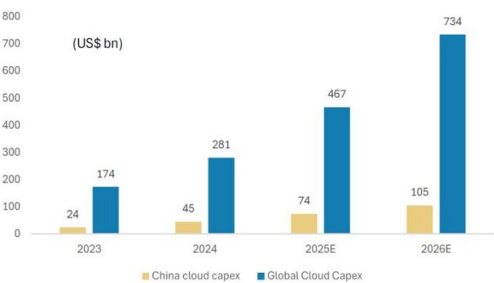


Surge in ByteDance (Volcano Engine/Doubao) tokens indicates high AI demand



Source: Company data, Taobao, Morgan Stanley Research.

China vs. global – cloud capex trend



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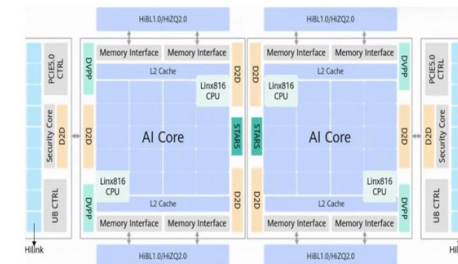
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AI Chip Value Chain – China vs. the US – A Decoupling of AI Computing

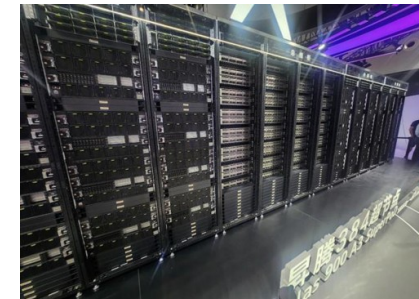


Source: Morgan Stanley Research.

Nine factors comparison between US and China AI – Chip, System, Infrastructure



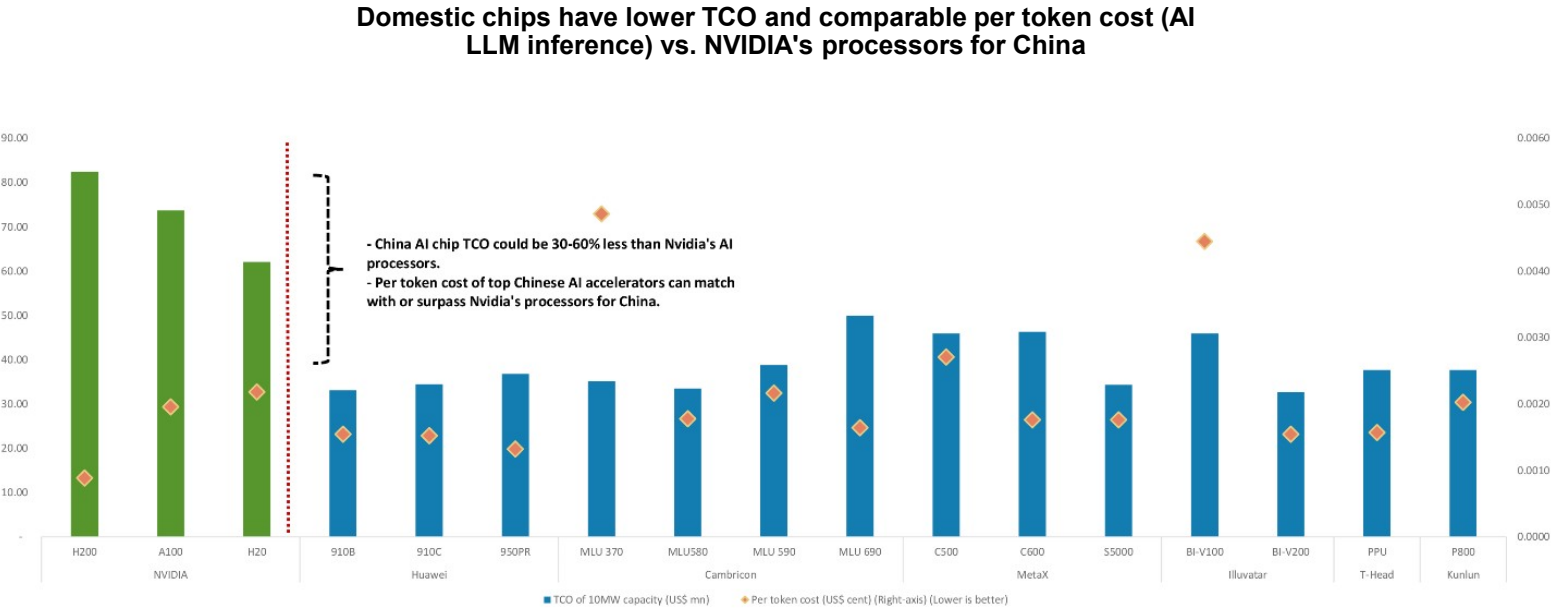
Huawei CloudMatrix 384 A3 SuperPod



Step 3: If one fab is not sufficient, expand manufacturing capacity.

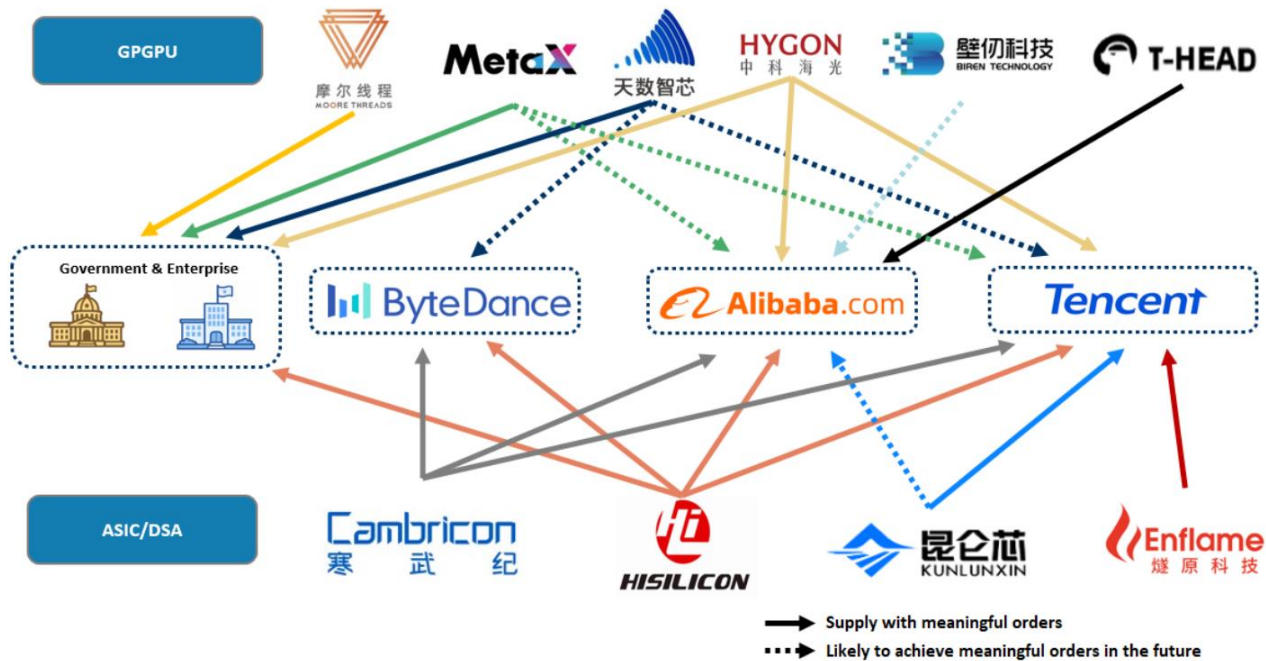
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Inference Economics: TCO and Cost per Token



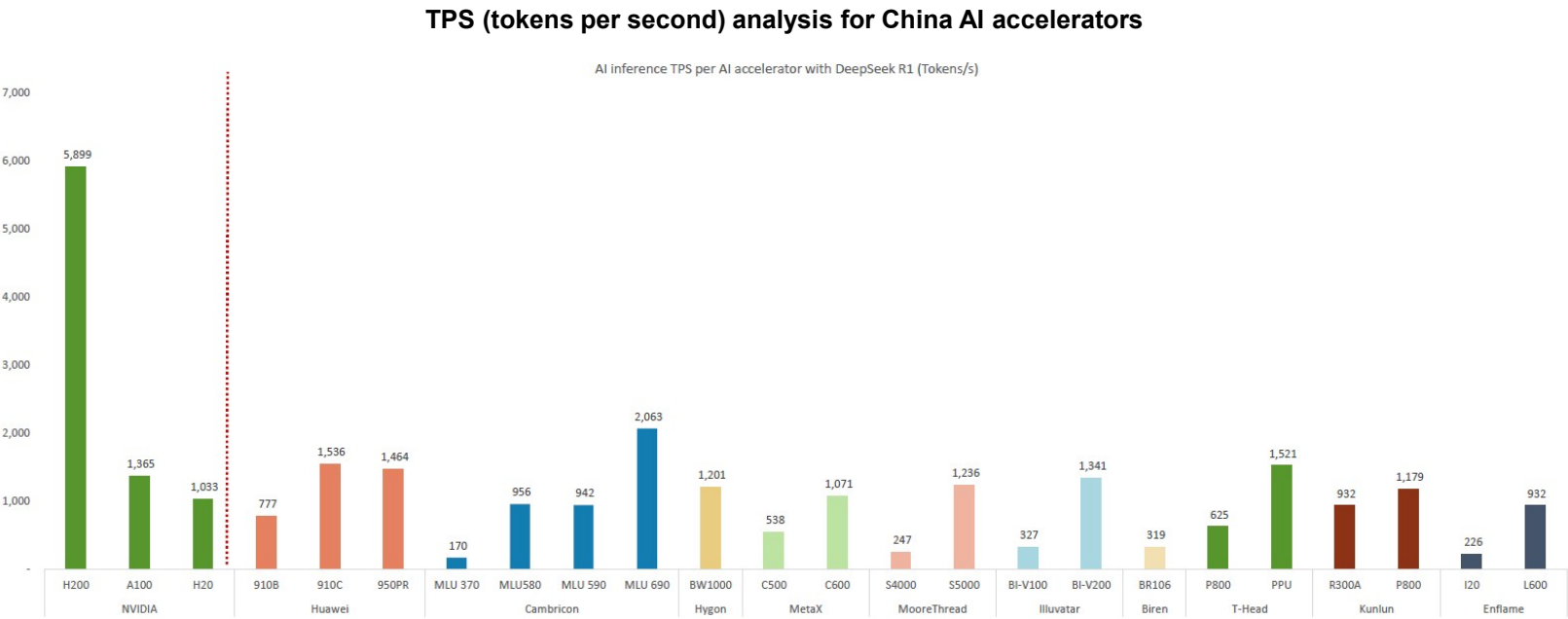
Source : Company data, Morgan Stanley Research estimates.

Order Placement and Potential Orders for Domestic AI Accelerator Developers



Source : Morgan Stanley Research

TPS (Tokens Output per Second) – Performance Analysis



Source: Company data, Morgan Stanley Research estimates (*Our assumptions include: AI LLM: DeepSeek R1; Input token: 1,024; Output token: 1,024; Number of active experts: 9 out of 257; Model size: 671GB; Model layer: 61; Batch size: 1; Computing power: FP8, and if FP8 is not available, we will use FP16).

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Domestic Chips Deliver Stronger Performance per Dollar Thanks to Materially Lower Pricing



* = Based on Morgan Stanley estimates. Source : Company data, Morgan Stanley Research.

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"10 Dragons" of Chinese AI GPGPU vendors. We Focus on Cambricon (OW), MetaX (EW), and Iluvatar (OW)

Company	Ticker	GPU Product	Node	GPGPU/ASIC	Foundry sourcing viability	Sovereign background	Affiliated design house	AI inference performance
 HUAWEI	Private	950, 910C, 910B	7nm	ASIC	✓	✓	✗	➡
 寒武纪	688256-SS	MLU series	7nm	ASIC	✓	✗	✗	➡
 HYGON	688041-SS	DCU	7nm	GPGPU	✓	✓	✗	➡
 沐曦	688802-SH	MXC series; MXG series	7nm/12nm	GPGPU	✓	✗	✗	➡
 摩尔线程 MOORE THREADS	688795-SH	MUSA MTT S5000, S4000	7nm/12nm	GPGPU	▲	✗	✗	➡
 壁仞科技 BIREN TECHNOLOGY	6082-HK	BR100	7nm/12nm	GPGPU	▲	✗	✗	➡
 天数智芯 Iluvatar CoreX	9903-HK	TianGai-100 series, Zhikai-100 Series	7nm	GPGPU	✓	✗	✗	➡
 T-HEAD	Currently under H-share listing counseling process	Hangguang 800, PPU	12nm	GPGPU	✓	✗	✓	➡
 昆仑芯 KUNLUNXIN	Currently under H-share listing counseling process	R and P series	6nm	ASIC	▲	✗	✓	➡
 燧原科技	Currently under A-share listing counseling process	S60, I20, L600	12nm	ASIC	▲	✗	✓	➡

Source: Company data, Morgan Stanley Research.

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Comparison among Cambricon, MetaX, and Iluvatar

			
Ticker	688256-SH	688802-SH	9903-HK
Product	MLU 220/270/370/580/590/690 (AI Training + inference)	C Series (AI training + inference) N Series (AI Inference) G Series (Graphic rendering)	Tiangai 100/150/200/300 (AI training + inference) Zhikai 100 (AI Inference)
GPGPU/ASIC	ASIC	GPGPU	GPGPU
Chip suppliers			
Process node for latest products	7nm/N+2	12nm/N+1	7nm
Secured orders from major CSPs	✓	✗	✓
Sovereign fund as major shareholder	✗	✓	✗
Per token cost performance	↑	→	→
2025 Revenue (Rmb mn)	CNY 6,497	CNY 1,644	CNY 1,034
Profitability	✓	✗	✗

Source: Company data, Morgan Stanley Research.

Cambricon: Leading in inference Performance and Customer Anchoring; OW

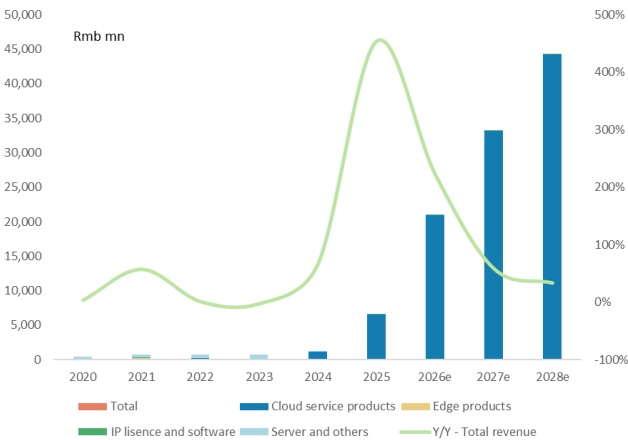
Cambricon’s products and specs

	Product	Type	Manufacture	Node	Computing power (TFLOPS/TOPS)			Memory type	Memory size (GB)	Memory bandwidth (GB/s)	Chip-to-chip networking bandwidth (GB/s)	TDP (W)
					FP16 (TFLOPS)	FP8 (TFLOPS)	Int8 (TOPS)					
AI Training	MLU 370	DSA	TSMC	7nm	96		X	X	LPDDR5	48	614	250
	MLU 580*	DSA	SMIC	12nm	280	560		X	HBM3	96	425	560
	MLU 590*	DSA	TSMC	7nm	315	630		X	HBM3	96	425	580
	MLU 690*	DSA	SMIC	7nm	700	1,400		X	HBM3	96	1,200	750
Edge AI	MLU 220	DSA	TSMC	16nm			X	16	LPDDR4	8	400	15

Cambricon’s supply chain and customers



We expect Cambricon's revenue to expand at a 90% CAGR, 2025-28e



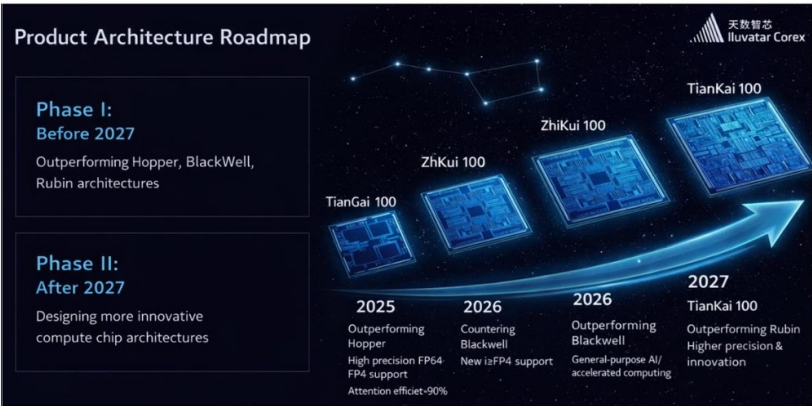
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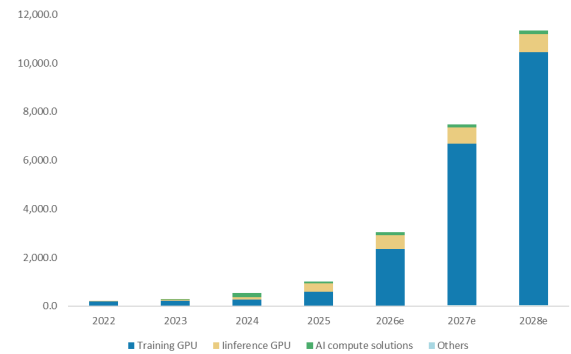
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Iluvatar: Leveraging Supply Chain Resilience with Strong Order Visibility; OW

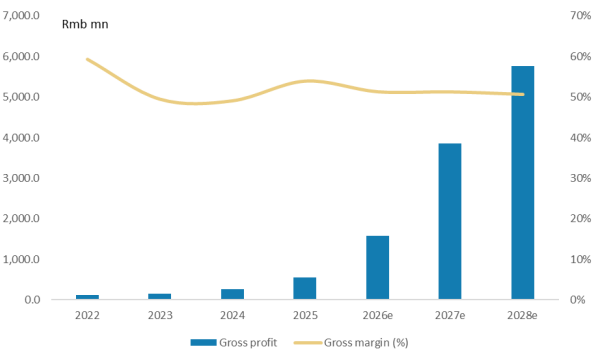
Aggressive product roadmap for Iluvatar



We expect Iluvatar's revenue to expand at a 122% CAGR, 2025-28e



We expect Iluvatar's GM to be mild



Source: Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates.

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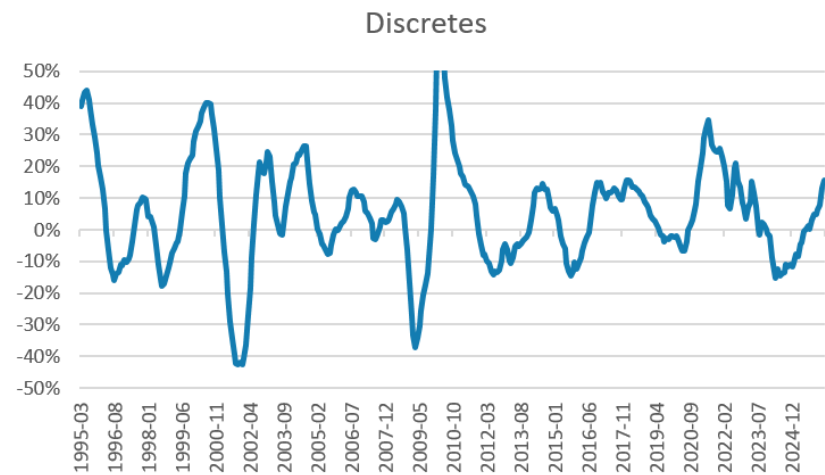
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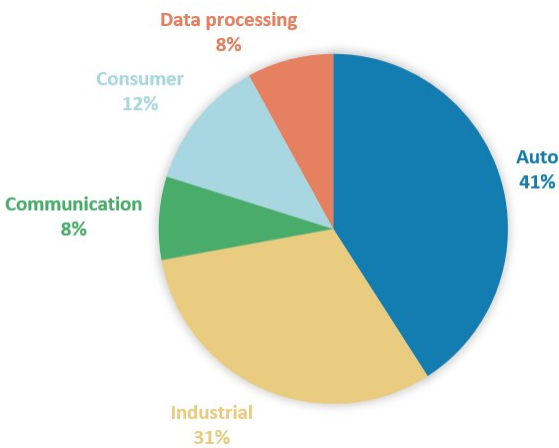
Power Semis – Supply Driven Upcycle

Total discrete revenue yoy growth turned positive since 4Q25



Source: WSTS, Gartner, Morgan Stanley Research.

Auto and industrial accounted for ~70% of end demand in discrete semiconductors (2025)

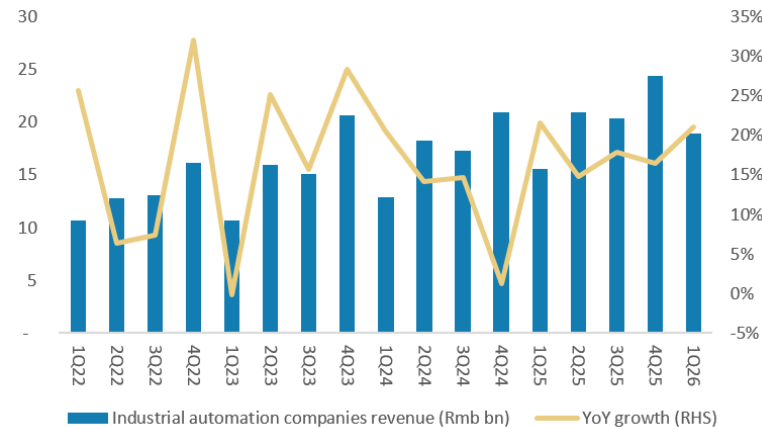


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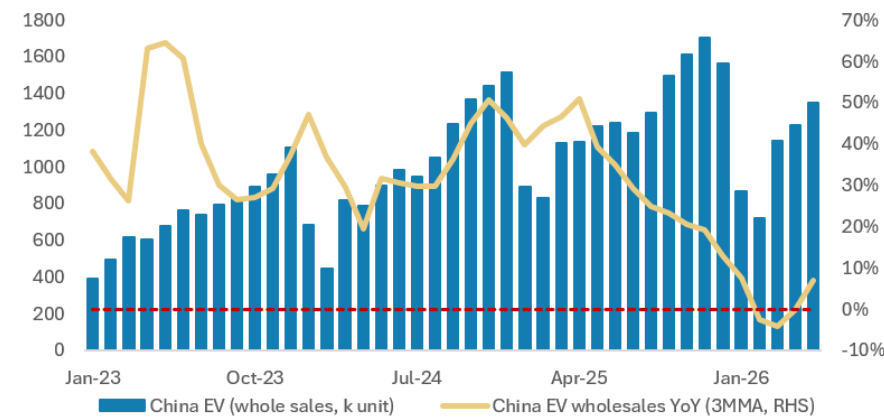
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Demand – Mixed Picture

Industrial automation companies' revenue
grew robustly at 21% YoY in 1Q26



China EV wholesale growth is weak YTD but turning
incrementally positive



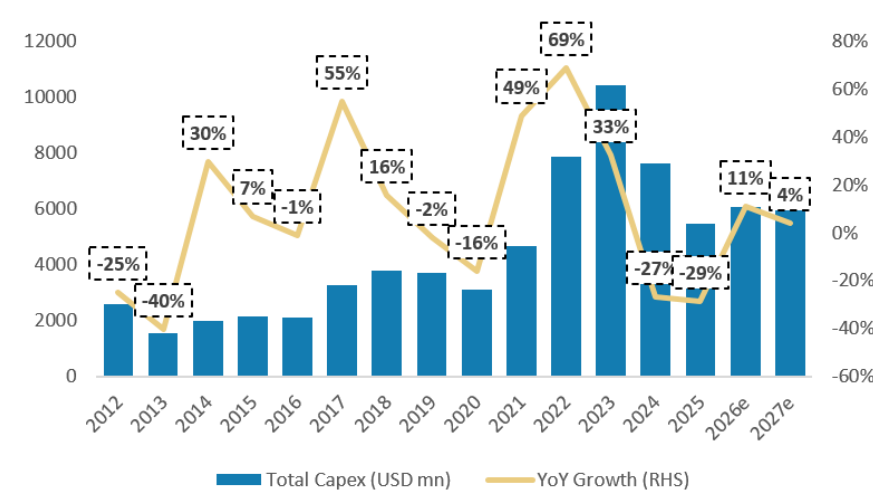
Source: CADA, Company data, Morgan Stanley Research.

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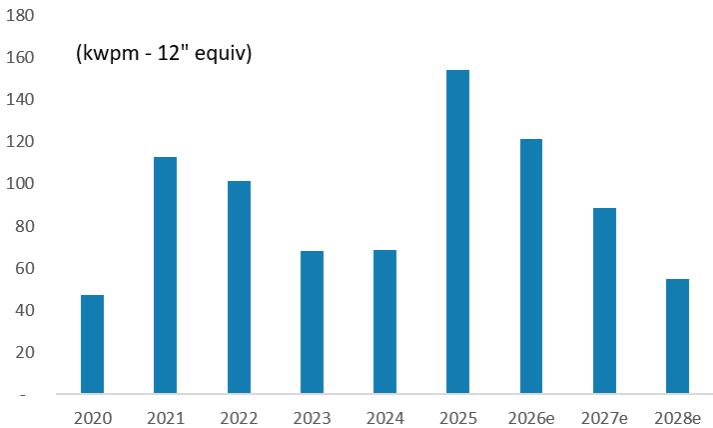
Limited Supply Addition

Global leading power semi companies' capex declined for 2 years



Source: Bloomberg, Morgan Stanley Research,
Note: estimates data using Bloomberg consensus numbers

Capacity additions likely to slow in 2026-28



Source: Company data, Morgan Stanley Research estimates

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Share price (last close)

Cambricon Technology Corporation (688256.SS)	1413 CNY
Iluvatar CoreX Semiconductor Co. Ltd (9903.HK)	735.50 HKD
MetaX Integrated Circuits (688802.SS)	768.98 CNY
Murata Manufacturing (6981.T)	11,005 JPY
Nvidia Corp. (NVDA.O)	208.65 USD
Samsung Electro-Mechanics (009150.KS)	1,990,000 KRW

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Global Stock Ratings Distribution

(as of May 31, 2026)

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Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1542	42%	465	51%	30%	707	43%
Equal-weight/Hold	1571	43%	369	40%	23%	723	44%
Not-Rated/Hold	3	0%	0	0%	0%	1	0%
Underweight/Sell	551	15%	86	9%	16%	201	12%
Total	3,667		920			1632	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

Analyst Stock Ratings

Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Equal-weight (E). The stock's total return is expected to be in line with the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Not-Rated (NR). Currently the analyst does not have adequate conviction about the stock's total return relative to the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Underweight (U). The stock's total return is expected to be below the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

Analyst Industry Views

Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.

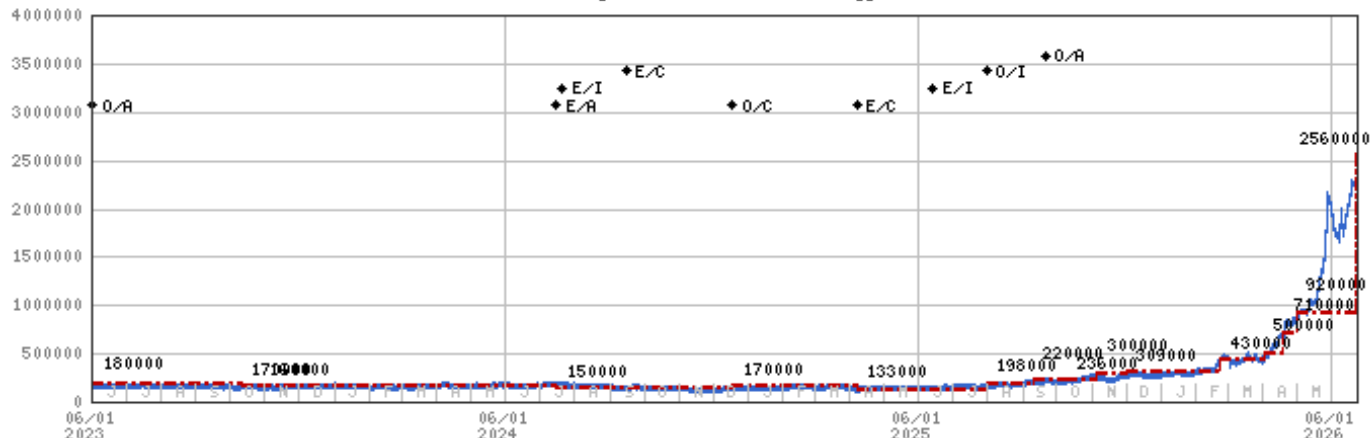
In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.

Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.

Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Stock Price, Price Target and Rating History (See Rating Definitions)

Samsung Electro-Mechanics (009150.KS) - As of 06/23/26 GMT in KRW
Industry : S. Korea Technology



Stock Rating History: 6/1/21 : E/A; 7/19/21 : E/I; 8/12/21 : E/C; 6/1/22 : U/C; 10/4/22 : U/A; 11/9/22 : O/A; 7/15/24 : E/A; 7/21/24 : E/I; 9/15/24 : E/C; 12/18/24 : O/C; 4/7/25 : E/C; 6/13/25 : E/I; 7/31/25 : O/I; 9/21/25 : O/A
Price Target History: 5/12/21 : 180000; 10/19/21 : 170000; 10/27/21 : 160000; 1/24/22 : 180000; 3/18/22 : 170000; 6/1/22 : 140000; 11/9/22 : 180000; 10/11/23 : 170000; 10/27/23 : 160000; 7/15/24 : 150000; 12/18/24 : 170000; 4/7/25 : 133000; 7/31/25 : 198000; 9/10/25 : 220000; 10/10/25 : 236000; 11/5/25 : 300000; 12/1/25 : 309000; 2/23/26 : 430000; 4/2/26 : 500000; 4/19/26 : 710000; 5/1/26 : 920000; 6/22/26 : 2560000

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target -- No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —
Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

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INDUSTRY COVERAGE: Technology - European Semiconductors

COMPANY (TICKER)	RATING (AS OF)	PRICE* (06/22/2026)
Lee Simpson		
ASML Holding NV (ASML.AS)	O (09/22/2025)	€1,655.80
Infineon Technologies AG (IFXGn.DE)	O (02/06/2025)	€85.94
STMicroelectronics NV (STMPA.PA)	O (03/26/2026)	€69.39
Nigel van Putten		
Aixtron SE (AIXGn.DE)	E (05/25/2023)	€60.42
ASM International NV (ASML.AS)	O (06/19/2024)	€1,079.00
BE Semiconductor Industries NV (BESI.AS)	O (11/07/2022)	€320.30
Melexis N.V. (MLXS.BR)	E (02/05/2026)	€85.00
Nordic Semiconductor ASA (NOD.OL)	E (02/10/2025)	NKr 191.40
Soitec SA (SOIT.PA)	O (03/26/2026)	€129.45
VAT Group AG (VACN.S)	E (03/21/2025)	SFr 706.80

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